FraudGuard AI — Handover Document for Member A

Layer 3: Autonomous Multi-Agent Investigation Pipeline (Architecture, Execution Flow & Integration)

ROLE / AUTHOR	TARGET RECIPIENT	SERVICE PORT / PROTOCOL	LLM RUNTIME / STATUS
Member C (Shruti Jaggi)	**Member A (UI / App Backend)**	**Port 8001 / HTTP & ASGI**	**Ollama llama3.2 / Operational**

1. Executive Overview & Scope Delivered

Member C has built and delivered the **Layer 3 Agentic Investigation Layer** for FraudGuard AI. While Member A operates the mock UPI client (Layer 1) and Member B maintains the GraphSAGE + XGBoost scoring engine (Layer 2), Member C's service provides the **autonomous multi-agent reasoning core**. When an account exceeds the operational risk threshold (blended score ≥ 0.85), this pipeline is triggered to gather multi-hop graph context, determine whether the pattern is a genuine fraud ring or an innocent decoy, enforce account freezes/clearances, and synthesize court-admissible forensic audit reports. Live benchmarking reveals a **13.12-second median flag-to-decision latency** (over **114× faster** than the 20–30 minute human analyst baseline).

2. Detailed Breakdown of the 4 Autonomous Agents

Agent & File	Forensic Core Responsibilities	Decisions & Output Signals
1. Investigator Agent nodes/investigator.py	• Analyzes account age, KYC tier, transaction velocity, retained balance, and 2-hop transaction neighbors. • **Discrimination Engine**: Distinguishes genuine fraud rings (*mule-chain, star-burst smurfing, cyclic wash trading, fan-out*) from innocent look-alike decoys (*family group, forwarding chain, merchant hub, payroll fanout*). • **Model Divergence Analysis**: Identifies structural over-flags where GraphSAGE (≥ 0.85) elevated an account due to graph proximity while tabular XGBoost (< 0.30) and tenure indicate an innocent false-positive.	Outputs: `suspicion_score` (0–100), `risk_level` (LOW/CRITICAL), topology booleans, `innocent_lookalike_detected`, and structured evidence trail.
2. Decision Agent nodes/decision.py	• **Stage 1 (Hard Business Rules Pre-Filter)**: Deterministic NPCI compliance enforcement. Auto-blocks confirmed mule-chains with rapid pass-through and near-zero retained balance; instantly clears verified decoys with tenure > 90 days and full biometric KYC. • **Stage 2 (LLM Forensic Judgment)**: Invokes local Ollama (`llama3.2:latest`) to evaluate ambiguous borderline cases where precision at threshold is ~0.43. • **Safety Guard**: Rule 3 (`BENIGN_BASELINE`) is strictly forbidden from clearing accounts with risk score ≥ 0.70 or victim/sink links.	Enforces one of 3 verdicts: **1. auto_block** **2. escalate** **3. clear** with confidence score & policy rationale.
3. Action Agent nodes/action.py	• **Pluggable Backends**: Operates in `SimulatedActionBackend` (safe local runs) or `MongoActionBackend` (direct persistence to MongoDB `InvestigationCase` and `AuditAction`). • **Action State Mapping**: Maps `auto_block` → `account_frozen` (revokes UPI handle, applies NPCI debit freeze, blacklists device ID); `escalate` → `flagged_for_review` (30-min hold, step-up OTP); `clear` → `cleared`. • **Score Invariance**: Never modifies Member B's precomputed ML scores.	Executes automated regulatory countermeasures, updates account status, and produces verified action summary strings.
4. Report Agent nodes/report.py	• Synthesizes executive, court-admissible forensic case audit reports detailing target account, KYC status, model score breakdown (Blended / XGBoost / GraphSAGE), detected topological patterns, forensic evidence trail, and policy rationales. • Appends standardized ISO 8601 audit timestamps and seals report with compliance metadata.	Returns sealed human-readable audit text formatted for both compliance logs and Member A's UI dashboard.

3. End-to-End Multi-Agent Pipeline Workflow

Trigger Event	Data Hydration	Agent 1: Investigate	Agent 2: Decide	Agent 3: Act	Agent 4: Report
Layer 2 model score ≥ 0.85 or UI manual challenge	Loads 893k profiles & 519k edges from CSV/Mongo	Discovers rings vs look-alikes; scores risk	NPCI rules pre-filter + Ollama LLM judgment	Freezes account, revokes VPA, blacklists device	Generates sealed forensic report for dashboard

4. Technology Stack Delivered

Component	Technology & Version	Architectural Rationale & Implementation Details
Web Service Core	`FastAPI (ASGI) / Uvicorn`	Async non-blocking REST endpoints, OpenAPI documentation at `/docs`, strict port 8001.
Agentic Framework	`LangGraph 0.2+ / LangChain`	Stateful DAG multi-agent execution pipeline passing validated `InvestigationState` dictionaries.
Reasoning Engine	`Ollama / llama3.2:latest`	Local 3.2B parameter LLM execution. 100% private, zero external cloud costs or API rate limits.
Schema Validation	`Pydantic v2`	Strict input/output JSON schema enforcement, JSON Schema structured output extraction from Ollama.
Data Provider Engine	`Dual-Mode CsvProvider & Mock`	Fast in-memory index over 893,216 account profiles and 519,454 transaction edges with sub-second lookups.
Persistence Layer	`MongoDB / PyMongo 4.0+`	Direct persistence into `InvestigationCase` and `AuditAction` collections.
Test & Quality Suite	`Pytest 9.0+ / Starlette TestClient`	26 automated tests covering contract shapes, thresholds, fallback caching, and business logic.

5. API Contract Specifications for Member A (Contract 4.1)

Member A's backend (`backend/services/agentServiceClient.js`) interacts with Layer 3 via a single, strongly-typed endpoint. The service guarantees **strict contract compliance**:

HTTP Method & Path: `POST http://127.0.0.1:8001/investigate`
Query Parameter: `?use_cache=true` (instant pre-cached response for live demo presentations)
Request Body Schema: `{"account_id": "C1000039615"}` (strips '@upi' and normalizes case)
Error Codes: `404 Not Found` for unknown accounts; `422 Unprocessable` for empty ID. *Never returns 500 on bad IDs.*

Success Response Body (200 OK):

```
{ "account_id": "C1000039615", "decision": "auto_block",
"action_taken": "UPI ID 'c1000039615@upi' revoked;
Immediate debit freeze applied on NPCI switch...",
"report": "=== FRAUDGUARD AI CASE AUDIT REPORT
===\nTarget Account: C1000039615..." }
```

6. Required Action Items & Integration Checklist for Member A

Item	Area in Member A Codebase	Required Implementation / Action	Status
1. Client Timeout	`backend/services/agentServiceClient.js`	Configure fetch timeout to ≥ **25–30 seconds** to allow complete local Ollama LLM reasoning without premature timeouts.	**Action Req.**
2. Payment Freeze Enforcement	`backend/routes/sendMoney.js`	When recipient account has `status === 'account_frozen'`, immediately reject transaction with debit freeze notification.	**Action Req.**
3. Socket.IO Live Feed	`frontend/src/components/AgentFeed.jsx`	Ensure UI listens to `io.emit('agent:activity')` to display real-time case reports and decision badges (Red/Amber/Green).	**Supported**
4. Account Status Enum	`backend/models/Account.js`	Standardized status strings: `account_frozen` (auto_block), `flagged_for_review` (escalate), `cleared` (clear).	**Aligned**
5. Demo Presentation IDs	Mock UPI App Pre-set Dropdown	**Mule Ring**: `C1000039615` (auto_block). **Innocent Decoy**: `acc_decoy_family_01` (resolved as clear false-alarm).	**Verified**

7. Quality Assurance & Sign-off

Verification Complete: 26 of 26 automated regression tests are passing (`pytest agent-service/test_pipeline.py`). The service runs continuously on `http://127.0.0.1:8001` with live CSV data ingestion active across all 893,216 dataset accounts. For interactive exploration, open the interactive Swagger UI documentation at **http://127.0.0.1:8001/docs**.